

Industrial Internship Report

Crop and Weed Detection using YOLOv8 & Turbofan Engine Remaining Useful Life Prediction

Prepared by

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Domain: Machine Learning / Data Science

upSkill Campus | The IoT Academy | UniConverge Technologies Pvt. Ltd. (UCT)

Executive Summary

This report presents the work completed during the Industrial Internship provided by upSkill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. (UCT). Two Machine Learning projects were completed: Crop and Weed Detection using YOLOv8, and Turbofan Engine Remaining Useful Life prediction using the NASA C-MAPSS FD001 dataset. The internship provided practical exposure to dataset preparation, model training, evaluation, GitHub documentation, and industry-style reporting.

1 Preface

This report summarizes the complete internship work. A relevant internship helps connect academic learning with industry-style problem solving, deadlines, and documentation.

The first project detects crops and weeds from field images using object detection. The second project estimates remaining useful life of turbofan engines from sensor data, which is useful for predictive maintenance.

upSkill Campus and UCT provided weekly learning material, project problem statements, quizzes, and a final GitHub submission process. I thank the mentors and management team of upSkill Campus, The IoT Academy, and UniConverge Technologies Pvt. Ltd. for this opportunity.

Message to juniors: start the project early, keep GitHub organized from day one, and write the report while doing the project.

2 Introduction

2.1 About UniConverge Technologies Pvt. Ltd.

UCT was established in 2013 and works in digital transformation. It provides industrial solutions using IoT, Cyber Security, Cloud Computing, Machine Learning, communication technologies, Java, Python, and frontend development. Relevant areas include IoT platforms, smart factory solutions, LoRaWAN applications, and predictive maintenance.

2.2 About upSkill Campus (USC)

upSkill Campus, along with The IoT Academy and UniConverge Technologies, facilitated this internship. It is a career development platform for upskilling, internships, projects, and industry exposure. Website: <https://www.upskillcampus.com/>

2.3 About The IoT Academy

The IoT Academy is the EdTech division of UCT and runs certification programs in collaboration with EICT Academies at IIT Kanpur, IIT Roorkee, and IIT Guwahati.

2.4 Objectives of this Internship Program

- Gain practical experience on industry-style problems
- Solve real-world Machine Learning tasks
- Improve job prospects through projects and GitHub
- Understand applications like computer vision and predictive maintenance
- Improve problem-solving and professional documentation

2.5 References

[1] Ultralytics YOLOv8 Documentation

[2] NASA C-MAPSS Turbofan Engine Degradation Dataset

[3] Scikit-learn and XGBoost documentation

[4] upSkill Campus internship guidelines and sample report format

[5] UniConverge Technologies company profile provided in internship material

2.6 Glossary

- YOLO: You Only Look Once, object detection algorithm
- YOLOv8: Ultralytics object detection model
- RUL: Remaining Useful Life
- C-MAPSS: Commercial Modular Aero-Propulsion System Simulation
- RMSE / MAE / R^2 : regression evaluation metrics
- mAP: Mean Average Precision
- UCT / USC: UniConverge Technologies / upSkill Campus

3 Problem Statement

Project 1: Crop and Weed Detection

Weeds compete with crops for nutrients, water, and sunlight. Manual identification is slow. The task was to detect and classify crop and weed regions in field images using bounding boxes.

Project 2: Turbofan Engine RUL Prediction

Machines degrade over time. If failure is not predicted early, downtime increases. The task was to predict Remaining Useful Life of turbofan engines using NASA C-MAPSS FD001 sensor data.

4 Existing and Proposed Solution

Existing crop-weed methods include manual inspection and simple image processing. These are less robust in real field conditions. The proposed solution uses YOLOv8 transfer learning with an organized train/validation dataset.

Existing RUL methods include scheduled maintenance and simple threshold alerts. The proposed solution uses RUL labeling, removal of constant sensors, scaling, and comparison of Random Forest, Gradient Boosting, and XGBoost.

4.1 Code submission (GitHub link)

<https://github.com/jnaneshm404/upskillcampus>

4.2 Report submission (GitHub link)

https://github.com/jnaneshm404/upskillcampus/blob/main/MLInternshipProjects_Inanesh_USC_UCT.pdf

5 Proposed Design / Model

5.1 Crop and Weed Detection flow

Input images → separate images and labels → train/val split → data.yaml → train YOLOv8n → evaluate → save weights and prediction results.

5.2 Turbofan RUL flow

Load FD001 files → name columns → calculate RUL → drop constant sensors → scale features → train models → compare RMSE, MAE, R^2 → save best model and plot.

5.3 Interfaces

Project 1: Colab/Jupyter + Ultralytics YOLO + data.yaml. Inference is on images, not live camera.

Project 2: Colab/Jupyter + Pandas, Scikit-learn, XGBoost, Matplotlib, joblib.

6 Performance Test

6.1 Test Plan

- Crop-Weed: check folder structure, data.yaml, training completion, sample inference
- RUL: check dataset loading, RUL calculation, metrics, saved model and plot

6.2 Test Procedure

YOLOv8n was trained on Google Colab GPU. Path issues between local system and Drive were fixed through data.yaml. RUL models were trained on processed FD001 features and compared using regression metrics.

6.3 Performance Outcome

Crop-Weed training completed and produced evaluation visuals and saved weights. Main constraints were Colab GPU limits and file-path errors.

RUL models were compared using RMSE, MAE, and R^2 . The best model and prediction plot were saved. Main constraints were constant-sensor removal and Colab path setup.

7 My Learnings

I learned YOLO dataset setup, object detection training, RUL labeling, preprocessing, model comparison, evaluation metrics, GitHub organization, and professional reporting. These skills will help in future internships, MCA project work, and AI/ML applications.

8 Future Work Scope

- Train a larger YOLO model or more epochs
- Add a simple interface to upload farm images
- Try LSTM or sequence models for RUL
- Deploy the best RUL model as an API
- Use a more diverse agricultural dataset